HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Architected and implemented **FastAPI** microservices for low-latency AI model inference, handling up to **~450 requests per second** during peak loads.
- Developed **Python** data pipelines on **AWS Lambda** and **S3** to ingest and pre-process terabytes of financial transaction data for model training.
- Integrated **LangChain** and **LlamaIndex** for building **RAG** pipelines, enabling LLMs to query internal knowledge bases and respond to complex queries.
- Designed and deployed **LLM** models within an **AWS** cloud environment, optimizing for performance and cost-efficiency using services like **SageMaker**.
- Collaborated with data scientists and product managers in an **agile** team to define technical requirements and deliver scalable AI solutions.
- Engineered **Java Spring Boot** services for seamless integration with existing financial systems, exposing robust **REST APIs** for data exchange.
- Containerized AI inference services using **Docker** and orchestrated deployments on **Kubernetes (EKS)**, ensuring high availability and efficient resource utilization.

SKILLS

AI/ML – Generative AI, LLMs, RAG, LangChain, LlamaIndex, Prompt Engineering, Model Deployment, Fine-tuning, Machine Learning

Cloud – AWS, EC2, S3, Lambda, SageMaker, CloudWatch, RDS, EKS

Programming Languages – Python, Java, SQL

Frameworks & Libraries – FastAPI, Flask, Spring Boot, scikit-learn, pandas, NumPy

Databases – PostgreSQL, Snowflake

DevOps & CI/CD – Docker, Kubernetes, Jenkins, Git, Terraform, Airflow

PROJECTS

Generative AI-Powered Financial Insights Platform

- Developed **RAG pipelines** using **LangChain** to allow an LLM to access and synthesize information from a large corpus of internal financial reports and market analysis documents.
- Built a **Python API** with **FastAPI** to serve LLM-generated insights, integrating with **PostgreSQL** for caching and session management.
- Fine-tuned a pre-trained transformer model on domain-specific financial language to improve response accuracy and reduce hallucination rates by **~18%**.

Machine Learning Model Deployment & Monitoring

- Engineered a **Python**-based feature pipeline on **AWS S3** and **Lambda** to prepare data for a fraud-detection model, processing over **~1.5 million** transaction records weekly.
- Deployed a trained **scikit-learn** classification model as a **REST API** using **Flask**, achieving a p95 inference latency of **220ms** on a dedicated EC2 instance.
- Implemented basic **model monitoring** in **CloudWatch** to track inference performance and data drift, triggering alerts for significant deviations.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)